First & Second Laws of Thermodynamics and Vapor Power Cycles

Comprehensive notes covering closed and open steady-flow energy equations, Clausius inequality, entropy generation, and reheat Rankine cycles.

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1. First and Second Laws of Thermodynamics

Steady Flow Energy Equation: h1 + (V1^2)/2 + gz1 + q = h2 + (V2^2)/2 + gz2 + w. Carnot maximum efficiency eta = 1 - (TL/TH).

2. Rankine Vapor Power Cycle

The standard cycle comprises pump compression, boiler heat addition, turbine expansion, and condenser rejection, enhanced by reheat.